currencies, etc. Derivatives can be used to manage various types of risks, such as price, interest rate, and credit risk, as well as to speculate on future price movements of the underlying asset. In this unit, we shall discuss various aspects of derivatives.

13.2 MEANING OF DERIVATIVES

In finance, a derivative is a financial instrument that derives its value from an underlying asset, such as a stock, bond, currency, or commodity. The value of a derivative is based on the changes in the value of the underlying asset. Simply put, if an asset, say A, is valued based on another asset, say B, which is linked to it, then A is called a derivative and B is called an underlying asset. The most common types of derivatives include *options, futures, forwards, and swaps*. Derivatives play an important role in financial markets, allowing investors and companies to manage risk and enhance returns, but they can also be complex and carry significant risk if not used properly.

The Securities Laws (Second Amendment) Act, 1999, has officially included derivatives in the Securities definition. According to the Securities Contracts (Regulations) Act, a derivative comprises securities derived from debt instruments, shares, loans, risk instruments, or contracts for differences. It also includes contracts deriving their value from the prices or index of underlying securities.

13.3 CHARACTERISTICS OF DERIVATIVES

The characteristics of derivatives are as follows:

a) A derivative derives its value from an underlying asset like a share or stock of a company. This feature makes it a unique financial asset.

b) Derivatives are traded globally and are very popular in developed economies.

c) Common derivatives include futures contracts, forward contracts, options, and swaps.

d) Most derivatives are not traded on exchanges and are used by institutions to hedge risk or speculate on price changes in the underlying asset.

e) Exchange-traded derivatives like futures or stock options are standardized and eliminate or reduce many of the risks of over-the-counter derivatives.

f) Derivatives are usually leveraged instruments, which increases their potential risks and rewards.

Advantages of Derivatives

a) **Lock in prices:** Derivatives help to protect the securities against fluctuations in prices. Thus, it helps to reduce market uncertainties. Companies and investors can hedge against price fluctuations in commodities, currencies, or securities, ensuring they are not adversely affected by market volatility. It also helps in financial planning and budgeting as the market uncertainty is reduced.

b) **Hedge against risk**: Derivatives are widely used to hedge against various types of risks, providing a safeguard for investors and businesses. Businesses involved in the production or consumption of commodities can use futures or options to lock in prices, protecting themselves from adverse price movements. Companies with debt or fixed-income investments can use interest rate swaps or futures to hedge against fluctuations in interest rates.

c) **Derivatives can be leveraged:** They allow investors to gain exposure to a larger position than what would be possible by investing directly in the underlying asset with the same amount of capital. Leverage can amplify returns on investment because a small change in the price of the underlying asset can lead to a larger change in the value of the derivative.

d) Useful in making a diversified portfolio: Derivatives can provide exposure to a wide range of asset classes, including commodities, currencies, interest rates, and equities, thus providing scope for diversification. Investors can also use derivatives to gain exposure to emerging markets, which might be otherwise inaccessible due to regulations or high entry costs.

Limitations of derivatives

a) **Difficult to value them:** Derivatives often require complex mathematical models to determine their fair value. For example, options pricing commonly uses the Black-Scholes model or binomial models, which involve advanced mathematics and assumptions about market behaviour. We will discuss Black-Scholes model and binomial pricing models in Unit 21.

b) **Subject to counterparty default:** If a counterparty defaults, the non-defaulting party may incur significant financial losses as the value of a derivative can be heavily dependent on the creditworthiness of the counterparty. A decline in the counterparty's credit rating can decrease the value of the derivative.

c) **Complex to understand:** There are numerous types of derivatives, such as options, futures, forwards, swaps, and more complex structures like collateralized debt obligations (CDOs) and credit default swaps (CDS). Each type has its own set of rules, characteristics, and uses. Moreover, pricing of derivatives also involves complex procedure.

d) **Sensitive to demand and supply factors**: The prices of derivatives can be highly volatile due to changes in demand and supply in the underlying asset markets. Moreover, high levels of speculative trading can amplify price movements and thus make derivatives volatile.

13.4 A BRIEF HISTORY OF DERIVATIVES IN INDIA

The historical roots of derivatives extend further back than commonly acknowledged, with some texts even suggesting their existence in incidents dating back to the Mahabharata. Traces of derivative contracts can be found in historical occurrences predating the era of Jesus Christ. Nevertheless, the formalization of modern derivative contracts is often attributed to the necessity for farmers to safeguard against potential declines in crop prices caused by delayed monsoons or overproduction.

India has a longstanding association with derivatives, particularly in the commodity market. Organized trading in cotton, dating back to 1875 with the establishment of the Cotton Trade Association, marked the initiation of the commodity derivative market in the country. Over the years, contracts for various other commodities have been introduced. In June 2000, exchange-traded financial derivatives were introduced at major stock exchanges in India, namely NSE and BSE. The National Commodity & Derivatives Exchange Limited (NCDEX) commenced operations in December 2003, providing a platform for commodities trading.

The derivatives market in India has witnessed substantial growth, especially at NSE. As of April 13, 2005, Stock Futures constitute the most actively traded contracts on the NSE, contributing to approximately 55% of the total turnover of derivatives.

Derivative trading in India can occur either on a distinct Derivative Exchange or within a designated segment of an existing Stock Exchange. These Derivative Exchanges/Segments operate as Self-Regulatory Organizations (SROs), with SEBI serving as the overseeing regulator. The clearing and settlement of all derivative trades on these platforms are facilitated through independent Clearing Corporations/Houses.

The introduction of derivative products in India occurred in stages, beginning with Index Futures Contracts in June 2000. Subsequently, Index Options and Stock Options were introduced in June and July 2001, respectively, followed by Stock Futures in November 2001. Derivatives trading in sectoral indices gained approval in December 2002. In June 2003, Interest Rate Futures on a notional bond and T-bill priced off ZCYC were introduced. Furthermore, exchange-traded interest rate futures on a notional bond priced off a basket of Government Securities received approval for trading in January 2004.

Check Your Progress 1

1) What do you mean by derivatives? Why are they called derivatives?

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2) What are the characteristics of derivatives?

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3) Discuss the advantages and disadvantages of derivatives.

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13.5 TYPES OF DERVIATIVES

Derivatives can be categorized based on various criteria, such as the nature of the underlying asset, trading venue, and contract structure. Although there are many different forms of derivatives, the following are some of the most popular ones in finance are:

1) **Futures:** Futures contracts are agreements to buy or sell an underlying asset at a predetermined price and time in the future. They are commonly used to hedge against price fluctuations and are traded on organized exchanges.

2) **Forwards:** Forwards are similar to futures, but they are not traded on exchanges. Instead, they are customized contracts between two parties to buy or sell an underlying asset at a predetermined price and time in the future.

3) **Options:** Options are contracts that give the buyer the right, but not the obligation, to buy or sell an underlying asset at a predetermined price and time. They can be used for hedging or speculative purposes, and are traded on both exchanges and over-the-counter markets.

4) **Swaps:** Swaps are agreements to exchange cash flows based on different financial instruments, such as interest rates, currencies, or commodities. They are commonly used to manage risk and are traded over the counter. The two most popular swaps are interest rate swaps and credit default swaps (CDS).

5) **Credit derivatives:** Credit derivatives are contracts that allow investors to manage credit risk by transferring it to another party. They include credit default swaps, which allow investors to protect themselves against

default on a debt obligation, and collateralized debt obligations (CDOs), which allow investors to pool together different types of debt securities.

6) **Equity derivatives:** Equity derivatives are contracts that allow investors to gain exposure to changes in the price of a stock or stock index. They include options, futures, and swaps on individual stocks or broad market indices.

7) **Commodity derivatives:** Commodity derivatives are contracts that allow investors to gain exposure to changes in the price of a commodity, such as gold, oil, or wheat. They include futures, options, and swaps on various commodities.

13.6 FUTURES AND FORWARDS

13.6.1 Meanings of Futures and Forwards

1) **Forwards** – A forward contract is a financial agreement between two parties where they agree to buy or sell an asset at a predetermined price at a future date. The parties involved in a forward contract can be individuals, corporations, or financial institutions. Forward contracts are used to hedge against price fluctuations in an asset, such as commodities, currencies, stocks, or bonds.

   Example 13.1: If a company wants to buy a certain commodity at a future date, it can enter into a forward contract with a supplier to buy the commodity at a fixed price, thus avoiding the risk of price volatility. Similarly, if an investor expects a certain stock to appreciate, they can enter into a forward contract to buy the stock at a predetermined price in the future.

2) **Futures** – Futures contracts are financial instruments that allow parties to buy or sell an asset, such as commodities, currencies, or financial instruments, at a predetermined price and date in the future. The price of the underlying asset is fixed at the time the futures contract is created, and the transaction is legally binding.

   Futures contracts are often used by traders and investors to hedge against potential price fluctuations or to speculate on the direction of market movements. For example, a farmer might sell a futures contract on a commodity like wheat to lock in a price for a future delivery, thereby reducing the risk of a price drop. Similarly, a speculator might buy a futures contract on a stock index if they believe that the market will rise in the future.

   We will discuss more about Futures and Forwards in Unit 21-Pricing of Derivatives.

13.6.2 Differences between Futures and Forwards

Futures and Forwards can be differentiated as follows:

Table 13.1: Differences between Futures and Forwards

S.No.	Forward contracts	Futures contracts
1	The contract price is not publicly disclosed	The contract price is transparent.
2	The contract is exposed to default risk by the counterparty.	The contract has effective safeguards against defaults.
3	Each contract is unique in terms of size, expiration date and asset type.	The contracts are standardized in terms of size, expiration date and all other futures.
4	Forward contracts suffer from the problem of liquidity.	There is no problem of liquidity.
5	Settlement of the contract is done by delivery of the asset on the expiration date.	Settlement is done on a cash basis.
6	They are traded on a personal basis or the telephone or otherwise.	They are traded in a competitive market.
7	They are traded in an over-the-counter market.	They are traded on organised exchanges with a designated physical location.
8	The costs of forward contracts are based on the bid-ask spread.	They involve brokerage frees.
9	Forward contracts are not subject to marking to market.	They are subject to marking to market.
10	Credit risk is borne by each party.	Credit risk is not borne by each party.

13.7 OPTIONS

An option is an agreement between two parties (say A and B) in which a party, say, A, has the right to buy or sell a particular asset at a predetermined price within a specified time frame. Moreover, if A does not want to execute his right, then he cannot be forced to do so by B. That is why the agreement is called option not obligation. An option helps in managing the risk of ups and downs of the price of an asset in which A is interested. Making a strategy to reduce or manage risk is called hedging. So, an option is a hedging tool.

Now a question arises i.e., **when should A use his right (technically known as exercising option)?** Before answering this question, we should understand two main types of options i.e.

1) Call Option

2) Put Option

13.7.1 Call Option

Suppose that the current price of a share of a company is ₹ 500. There is a good chance that the price will go up in the next 4 months, but a fall is also probable in the price. You are in a dilemma to buy the share or not. Luckily, you find a person, say, Vikas, who agrees to give you an option to buy the share from him at ₹ 530 after 4 months irrespective of the actual price of the share in the market. For this, Vikas will charge a cost of ₹ 20 as purchase cost. Common sense says that if the actual price of the share goes up, say, ₹ 580, then it will be profitable for you to purchase the share from Vikas at ₹ 530 and sell in the market at ₹ 580. It will create a profit of ₹ 30 after charging the cost of ₹ 20. Of course, it will be a loss-making situation for Vikas. Now, what if the price does not reach the level which creates profit for you? In that case, you cannot be forced to buy the share from him because it is an option, not an obligation. Thus, by paying an amount of ₹ 20 only, you can avail the chance or opportunity of profit with zero risk or you can hedge your risk.

In finance, such an agreement is called a **call option**, the share of the company is called **underlying asset**, the price demanded by Vikas is called the **exercise price or strike price**, and the difference between the exercise price at expiry and the price of the underlying asset i.e., ₹ 50 is called exercise value (or intrinsic value of the call option), the cost of ₹ 20 charged by him is called **option premium**, the profit of ₹ 30 earned is called net **payoff**, and Vikas is called **option writer** and you are an **option buyer**. Since this call option derives its value because of a share linked to it, therefore, it is a derivative also. Thus, we can define a call option as follows:

A **call option** is a type of derivative in which the buyer gets a right to **buy** an underlying at an agreed price (known as exercise price or strike price) when the buy is profitable to him at the time of expiry of the option. Since this derivative is an option, therefore, he cannot be forced by the option writer to buy the underlying asset when its buy is unprofitable, but for this option, he has to bear a cost in the form of option premium charged due to time value of money and risk taken by the option writer.

Now, we can answer the question, **when should A use his right (technically known as exercising option)?** The decision to exercise a call option can be taken as follows:

Table 13.2: Decision to exercise a call option

Situations	**Decision**	**Market Terms**
Price of the underlying asset (S) at option expiry > Exercise price (E) i.e., S – E > 0	Exercise the call option because it is a profitable situation for the buyer of the call option.	In-the-money call option i.e., money will come to the option buyer.

Price of the underlying asset (S) at option expiry < Exercise price (E) i.e., S – E < 0	Don't exercise the call option because it is not a profitable situation.	Out-the-money call option i.e., money would flow out if the option is exercised.

Thus, a call option should be exercised when S > E or S – E > 0. The *exercise value or intrinsic value* (or just value denoted by VCO) of a call option at its expiry can be calculated using the following formula:

$$\mathbf{VCO} = \begin{cases} \mathbf{S - E} & \mathbf{if\ S > E} \\ \mathbf{0} & \mathbf{if\ S \leq E} \end{cases}$$

A call option is exercised only when **S > E** because if a call option is exercised when **S < E,** then it will not be a rational decision because in that case receiving price S from the market by paying a higher amount E to the option writer is a loss-making transaction. Therefore, the value of the call option can also be expressed as this

VCO = Max [S – E, 0]

Max [S-E, 0] is read as chose the maximum of (S – E) and 0.

Profit on a call option is calculated using the following formula:

Net Payoff = VCO – option premium

If we plot the net payoffs against the prices of the underlying asset, then we get a graph known as the net payoff diagram. One point should be noted this diagram is drawn from **the option buyer's point of view**.

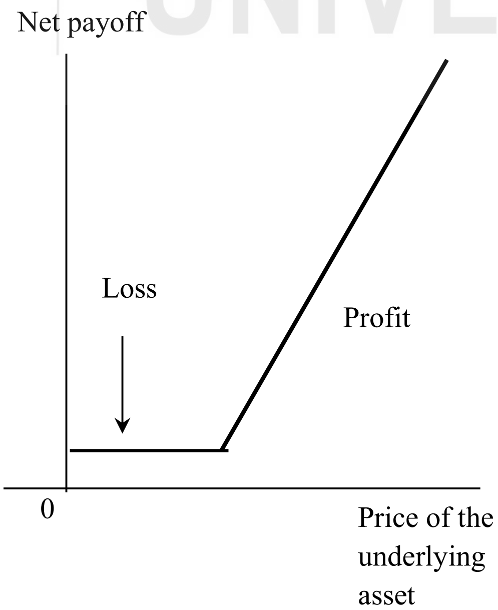


Fig. 13.1: Net payoff from the option buyer's point of view

If the net payoff diagram is drawn from the option writer's point of view, then the diagram is as follows:

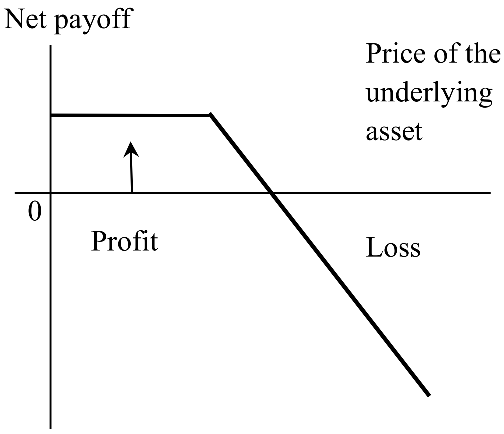


Fig. 13.2: Net payoff from the option writer's point of view

Let us understand Call option with the help of an example.

Example 13.2: Mr. Aakash is willing to buy a share of Reliance Industries and after 3 months, its probable prices are as follows: ₹ 5, ₹ 10, ₹ 15, ₹ 20, ₹ 25, ₹30. The current price is ₹ 7. Aakash purchases a call option from a derivative dealer and as per the terms and conditions, the exercise price is ₹ 12, and the option premium is ₹ 35.

a) You are required to suggest when to exercise the above call option.

b) Also, sketch the net payoff diagram from Aakash and the derivative dealer's point of view.

Solution

Net payoff call option buyer point of view i.e., Aakash.

Price of underlying asset (in ₹)	Option premium (in ₹)	Exercise price (in ₹)	Total cost (₹)	Net pay-off (₹)	Option exercise (Yes or No)
5	2	-	2	-2	No
10	2	-	2	-2	No
15	2	13	15	0	Yes
20	2	13	15	5	Yes
25	2	13	15	10	Yes
30	2	13	15	15	Yes

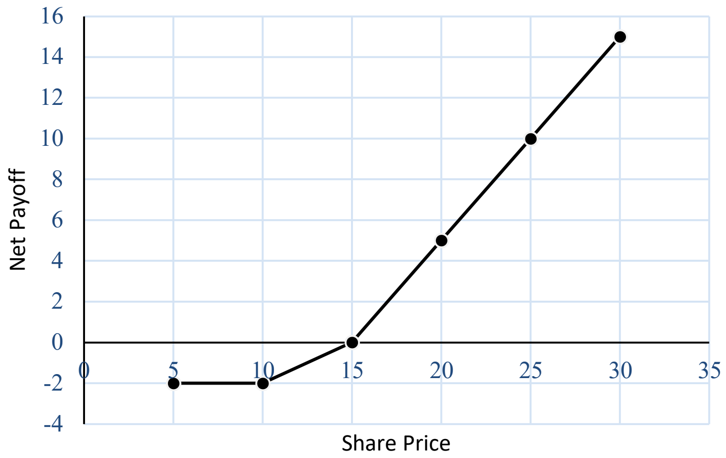


Fig. 13.3: Net Payoff Diagram from Akash's Point of View

Net payoff option writer point of view i.e., derivative dealer.

Price of underlying asset (in ₹)	Option premium (in ₹)	Exercise price (in ₹)	Total Revenue (₹)	Net payoff (₹)	Option exercise (Yes or No)
5	2	-	2	2	No
10	2	-	2	2	No
15	2	13	15	0	Yes
20	2	13	15	-5	Yes
25	2	13	15	-10	Yes
30	2	13	15	-15	Yes

Fig. 13.4: Net Payoff Diagram from option Writer's point of view

13.7.2 Put Option

A **put option** is a type of derivative in which the buyer gets a right to sell an underlying at an agreed price (known as exercise price or strike price) when the sale is profitable to him. Since this derivative is an option, therefore, he cannot be forced to sell the underlying asset when its sale is unprofitable, but for this option, he has to bear a cost in the form of an option premium. A put option can be exercised as follows:

Table 13.3: Decision to exercise a Put option

Situations	**Decision**	**Market Terms**
Exercise price (E) < Price of the underlying asset (S) at option expiry i.e., E– S < 0	Don't exercise the put option because it is a profitable situation for the buyer of the call option.	Out-the-money put option i.e., money will come to the option buyer.
Exercise price (E) > Price of the underlying asset (S) at option expiry i.e., E– S > 0	Exercise put option because it is not a profitable situation.	In-the-money put option i.e., money will come to the option buyer.

Thus, a put option should be exercised when E > S or E – S > 0. The *exercise value or intrinsic value* (or just value denoted by VPO) of a put option at its expiry can be calculated using the following formula:

$$\text{VPO} = \begin{cases} E - S & \text{if } E > S \\ 0 & \text{if } E \leq S \end{cases}$$

A put option is exercised only when E > S because if a put option is exercised when E < S, then it will not be a rational decision because receiving amount E from the option writer by purchasing for a higher amount S in the market is a loss-making transaction. Thus, the value of a put option can also be expressed as this

VPO = Max [E – S, 0]

Max [E-S, 0] is read as chose the maximum of (E – S) and 0.

Profit on a put option is calculated using the following formula:

Net Payoff = VPO – option premium

Example 13.3: Ms. Siddhi is willing to sell the share of SBI and after 2 months, its probable prices are as follows: ₹ 5, ₹ 10, ₹ 15, ₹ 20, ₹ 25, ₹30. The current price is ₹ 7. She purchases a put option from a derivative dealer and as per the terms and conditions, the exercise price is ₹ 12, and the option premium is ₹ 35.

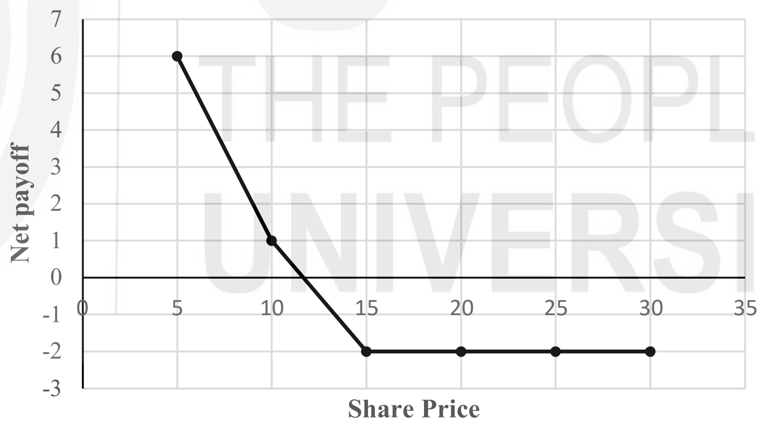
a) You are required to suggest when to exercise the above-put option.

b) Also, sketch the net payoff diagram from Siddhi and the derivative dealer's point of view.

Solution

Net payoff put option buyer point of view i.e., Ms Siddhi.

Price of underlying asset (in ₹)	Option premium (in ₹)	Exercise price (in ₹)	Total cost (₹)	Net pay-off (₹)	Option exercise (Yes or No)
5	2	13	7	6	Yes
10	2	13	12	1	Yes
15	2	-	2	-2	No
20	2	-	2	-2	No
25	2	-	2	-2	No
30	2	-	2	-2	No



Check Your Progress 2

1) What is an option? Why is it called an option?

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2) What are call and put options? Explain using an example.

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